

Geometry for the Curved-Space Dirac Equation

Coordinates

$$(t, r, \theta, \phi)$$

Line Element

$$ds^2 = F(r) dt^2 - \frac{1}{F(r)} dr^2 - r^2 d\theta^2 - r^2 \sin^2 \theta d\phi^2.$$

Metric Tensor

$$g_{\mu\nu} = \begin{pmatrix} F(r) & 0 & 0 & 0 \\ 0 & -\frac{1}{F(r)} & 0 & 0 \\ 0 & 0 & -r^2 & 0 \\ 0 & 0 & 0 & -r^2 \sin^2(\theta) \end{pmatrix}$$

Inverse Metric

$$g^{\mu\nu} = \begin{pmatrix} \frac{1}{F(r)} & 0 & 0 & 0 \\ 0 & -F(r) & 0 & 0 \\ 0 & 0 & -\frac{1}{r^2} & 0 \\ 0 & 0 & 0 & -\frac{1}{r^2 \sin^2(\theta)} \end{pmatrix}$$

Christoffel Symbols

The Christoffel symbols are defined by

$$\Gamma^\rho_{\mu\nu} = \frac{1}{2} g^{\rho\sigma} (\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}).$$

$$\Gamma^0_{01} = \frac{\frac{d}{dr} F(r)}{2F(r)}$$

$$\Gamma^1_{00} = \frac{F(r) \frac{d}{dr} F(r)}{2}$$

$$\Gamma^1_{11} = -\frac{\frac{d}{dr} F(r)}{2F(r)}$$

$$\Gamma^1_{22} = -rF(r)$$

$$\Gamma_{33}^1 = -rF(r)\sin^2(\theta)$$

$$\Gamma_{12}^2 = \frac{1}{r}$$

$$\Gamma_{33}^2 = -\frac{\sin(2\theta)}{2}$$

$$\Gamma_{13}^3 = \frac{1}{r}$$

$$\Gamma_{23}^3 = \frac{1}{\tan(\theta)}$$

Tetrad

The tetrad is obtained from

$$g_{\mu\nu} = \eta_{ab}e^a{}_{\mu}e^b{}_{\nu}, \quad \eta_{ab} = \text{diag}(1, -1, -1, -1).$$

$$e^a{}_{\mu} = \begin{pmatrix} \sqrt{F(r)} & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{F(r)}} & 0 & 0 \\ 0 & 0 & r & 0 \\ 0 & 0 & 0 & r \sin(\theta) \end{pmatrix}$$

Inverse Tetrad

$$e^a{}_{\mu}e^{\mu}{}_b = \delta^a_b, \quad e^a{}_{\mu}e^{\nu}{}_a = \delta^{\nu}{}_{\mu}.$$

$$e^{\mu}{}_a = \begin{pmatrix} \frac{1}{\sqrt{F(r)}} & 0 & 0 & 0 \\ 0 & \sqrt{F(r)} & 0 & 0 \\ 0 & 0 & \frac{1}{r} & 0 \\ 0 & 0 & 0 & \frac{1}{r \sin(\theta)} \end{pmatrix}$$

Spin Connection

The spin connection coefficients are

$$\omega_{\mu}{}^{ab} = e^a{}_{\nu} \left(\partial_{\mu} e^{b\nu} + \Gamma^{\nu}{}_{\mu\lambda} e^{b\lambda} \right).$$

Only the independent components with $a < b$ are displayed.

$$\omega_0^{01} = \frac{\frac{d}{dr}F(r)}{2}$$

$$\omega_2^{12} = -\sqrt{F(r)}$$

$$\omega_3^{13} = -\sqrt{F(r)} \sin(\theta)$$

$$\omega_3^{23} = -\cos(\theta)$$

Spinor Connection

$$\Gamma_{\mu} = \frac{1}{4} \omega_{\mu ab} \gamma^a \gamma^b.$$

$$\Gamma_0 = \begin{pmatrix} 0 & 0 & 0 & -\frac{\frac{d}{dr}F(r)}{8} \\ 0 & 0 & -\frac{\frac{d}{dr}F(r)}{8} & 0 \\ 0 & -\frac{\frac{d}{dr}F(r)}{8} & 0 & 0 \\ -\frac{\frac{d}{dr}F(r)}{8} & 0 & 0 & 0 \end{pmatrix}$$

$$\Gamma_1 = 0$$

$$\Gamma_2 = \begin{pmatrix} \frac{i\sqrt{F(r)}}{4} & 0 & 0 & 0 \\ 0 & -\frac{i\sqrt{F(r)}}{4} & 0 & 0 \\ 0 & 0 & \frac{i\sqrt{F(r)}}{4} & 0 \\ 0 & 0 & 0 & -\frac{i\sqrt{F(r)}}{4} \end{pmatrix}$$

Gamma3

$$\begin{pmatrix} 0 & \frac{i\left(i\sqrt{F(r)}\sin(\theta)+\cos(\theta)\right)}{4} & 0 & 0 \\ \frac{i\left(-i\sqrt{F(r)}\sin(\theta)+\cos(\theta)\right)}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i\left(i\sqrt{F(r)}\sin(\theta)+\cos(\theta)\right)}{4} \\ 0 & 0 & \frac{i\left(-i\sqrt{F(r)}\sin(\theta)+\cos(\theta)\right)}{4} & 0 \end{pmatrix}$$

Flat-Space Gamma Matrices

The flat-space gamma matrices satisfy

$$\{\gamma^a, \gamma^b\} = 2\eta^{ab}I_4.$$

$$\gamma^0 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$\gamma^1 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}$$

$$\gamma^2 = \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \\ 0 & i & 0 & 0 \\ -i & 0 & 0 & 0 \end{pmatrix}$$

$$\gamma^3 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

Curved-Space Gamma Matrices

The curved-space gamma matrices are

$$\gamma^\mu(x) = e^\mu_a(x)\gamma^a.$$

$$\gamma^0(x) = \begin{pmatrix} \frac{1}{\sqrt{F(r)}} & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{F(r)}} & 0 & 0 \\ 0 & 0 & -\frac{1}{\sqrt{F(r)}} & 0 \\ 0 & 0 & 0 & -\frac{1}{\sqrt{F(r)}} \end{pmatrix}$$

$$\gamma^1(x) = \begin{pmatrix} 0 & 0 & 0 & \sqrt{F(r)} \\ 0 & 0 & \sqrt{F(r)} & 0 \\ 0 & -\sqrt{F(r)} & 0 & 0 \\ -\sqrt{F(r)} & 0 & 0 & 0 \end{pmatrix}$$

$$\gamma^2(x) = \begin{pmatrix} 0 & 0 & 0 & -\frac{i}{r} \\ 0 & 0 & \frac{i}{r} & 0 \\ 0 & \frac{i}{r} & 0 & 0 \\ -\frac{i}{r} & 0 & 0 & 0 \end{pmatrix}$$

$$\gamma^3(x) = \begin{pmatrix} 0 & 0 & \frac{1}{r \sin(\theta)} & 0 \\ 0 & 0 & 0 & -\frac{1}{r \sin(\theta)} \\ -\frac{1}{r \sin(\theta)} & 0 & 0 & 0 \\ 0 & \frac{1}{r \sin(\theta)} & 0 & 0 \end{pmatrix}$$

Covariant Derivative

For a Dirac spinor,

$$D_\mu \psi = (\partial_\mu + \Gamma_\mu) \psi.$$

$$D_0 \psi = \begin{pmatrix} -\frac{\psi_4(t, r, \theta, \phi) \frac{d}{dr} F(r) - 8 \frac{\partial}{\partial t} \psi_1(t, r, \theta, \phi)}{8} \\ -\frac{\psi_3(t, r, \theta, \phi) \frac{d}{dr} F(r) - 8 \frac{\partial}{\partial t} \psi_2(t, r, \theta, \phi)}{8} \\ -\frac{\psi_2(t, r, \theta, \phi) \frac{d}{dr} F(r) - 8 \frac{\partial}{\partial t} \psi_3(t, r, \theta, \phi)}{8} \\ -\frac{\psi_1(t, r, \theta, \phi) \frac{d}{dr} F(r) - 8 \frac{\partial}{\partial t} \psi_4(t, r, \theta, \phi)}{8} \end{pmatrix}$$

$$D_1 \psi = \begin{pmatrix} \frac{\partial}{\partial r} \psi_1(t, r, \theta, \phi) \\ \frac{\partial}{\partial r} \psi_2(t, r, \theta, \phi) \\ \frac{\partial}{\partial r} \psi_3(t, r, \theta, \phi) \\ \frac{\partial}{\partial r} \psi_4(t, r, \theta, \phi) \end{pmatrix}$$

$$D_2 \psi = \begin{pmatrix} \frac{i\sqrt{F(r)}\psi_1(t, r, \theta, \phi) + 4 \frac{\partial}{\partial \theta} \psi_1(t, r, \theta, \phi)}{4} \\ \frac{-i\sqrt{F(r)}\psi_2(t, r, \theta, \phi) + 4 \frac{\partial}{\partial \theta} \psi_2(t, r, \theta, \phi)}{4} \\ \frac{i\sqrt{F(r)}\psi_3(t, r, \theta, \phi) + 4 \frac{\partial}{\partial \theta} \psi_3(t, r, \theta, \phi)}{4} \\ \frac{-i\sqrt{F(r)}\psi_4(t, r, \theta, \phi) + 4 \frac{\partial}{\partial \theta} \psi_4(t, r, \theta, \phi)}{4} \end{pmatrix}$$

$$D_3 \psi = \begin{pmatrix} \frac{-\sqrt{F(r)}\psi_2(t, r, \theta, \phi) \sin(\theta) + i\psi_2(t, r, \theta, \phi) \cos(\theta) + 4 \frac{\partial}{\partial \phi} \psi_1(t, r, \theta, \phi)}{4} \\ \frac{\sqrt{F(r)}\psi_1(t, r, \theta, \phi) \sin(\theta) + i\psi_1(t, r, \theta, \phi) \cos(\theta) + 4 \frac{\partial}{\partial \phi} \psi_2(t, r, \theta, \phi)}{4} \\ \frac{-\sqrt{F(r)}\psi_4(t, r, \theta, \phi) \sin(\theta) + i\psi_4(t, r, \theta, \phi) \cos(\theta) + 4 \frac{\partial}{\partial \phi} \psi_3(t, r, \theta, \phi)}{4} \\ \frac{\sqrt{F(r)}\psi_3(t, r, \theta, \phi) \sin(\theta) + i\psi_3(t, r, \theta, \phi) \cos(\theta) + 4 \frac{\partial}{\partial \phi} \psi_4(t, r, \theta, \phi)}{4} \end{pmatrix}$$

Curved-Space Dirac Equation

The curved-space Dirac equation is

$$\boxed{[i\gamma^\mu(x) (\partial_\mu + \Gamma_\mu) - m] \psi(x) = 0}$$